Project 4 Report

By Emberly Walburger and Jordan Caldwell

Part 1

Introduction

For part 1 of this assignment, we analyzed the Cleveland heart disease dataset that was provided. In the Cleveland dataset, out of 303 total patients, 139 patients had heart disease, which is 45.87 %. Our goal was to create an optimal model for predicting if a patient has heart disease based on different attributes in the dataset, along with an optimal value of k for a K Nearest Neighbors analysis. Creating optimized models to predict things like heart disease is very important for medical research into correlation and causation. It is also important for informing patients and providers of risk factors to watch out for. Using four different attributes along with a k-value of 9 gave us the best predictive model. To see our analysis process and code, visit our github repository, ewalburger/CS5830_Project4. To see our presentation, view our google slides, Project 4 - k-nearest neighbors BY Jordan Caldwell and Emberly Walburger .

Methods

We mostly found a value of k and what attributes to use for k-nearest neighbors through trial and error. To find the ideal value of k, we created a loop, so as we looked for the best attributes, we would run each iteration through several different values of k first to try and figure out which one was best. We only used odd values of k, to avoid ties for the nearest neighbors. Originally, we tried some higher values of k, but it didn't seem to increase the F1, Precision, and Recall scores by going beyond 9 values of k, so we mainly just iterated through k = 3, 5, 7, 9. Eventually, we noticed a trend that 9 was always the highest overall F1, so we decided to exclusively use it for Part 1.

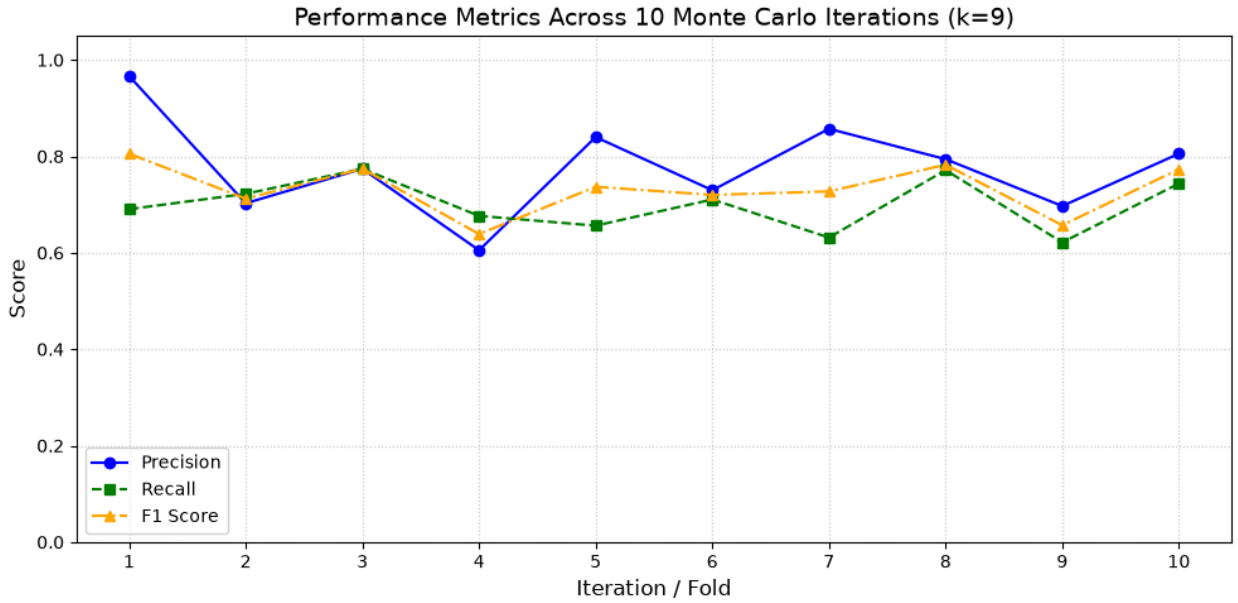
To find an optimal set of attributes, we also mostly tried meaningful trial and error. We researched what kinds of attributes are most meaningful for predicting the presence of heart disease, so we used that as a guide. We went through many iterations of trying to raise the F1, Precision, and Recall scores using different attributes. One big finding we had was that age is reportedly very important for predicting heart disease, but each time we used it, it lowered the scores. Same with 'trestbps'. The highest consistent scores that we got were with the combination of 'chol', 'oldpeak', 'cp', and 'thalach'. These values made sense to us, because they are some of the biggest predictors of heart disease.

Results

After many iterations of optimizing our k-value and attributes, we chose to use k=9 and an attribute combination of 'chol', 'oldpeak', 'cp', and 'thalach'. Over ten iterations of Monte Carlo

Cross Validation, we found a mean precision of 0.777, a mean recall of 0.700 and a mean F1 score of 0.733. This means that we were able to accurately predict the presence of heart disease 77.7 % of the time, and we correctly identified 70% of the people who had heart disease.

Overall, this analysis could be very useful to hospitals and doctors to give them a good idea of if their patient may have heart disease. It is also very useful in showing what factors can accurately predict heart disease. This can empower people to take measures such as watching their cholesterol, and can inform people of what kinds of chest pain to watch out for. This predictive model is not perfect, and is limited. If you were to only use this model, you would misdiagnose 30% of people who have heart disease as not having heart disease, which would have deadly consequences. Models can only tell you so much, which is why it is important for doctors to evaluate each patient on an individual level.



Part 2

Introduction

When we first started part 2 we had no idea of what to do the project on so we started to look for data sets on the website the teacher provided. Which is when we quickly came across an interesting data set about two Portuguese schools. The idea was to see if we could predict if students would pass or fail a math class based on attributes of the student schools. I think school systems could use this data, because if they know what attributes most strongly promote academic performance then they can do things to prompt those attributes. An example is if study time is important than they can offer after school programs to help kids study.

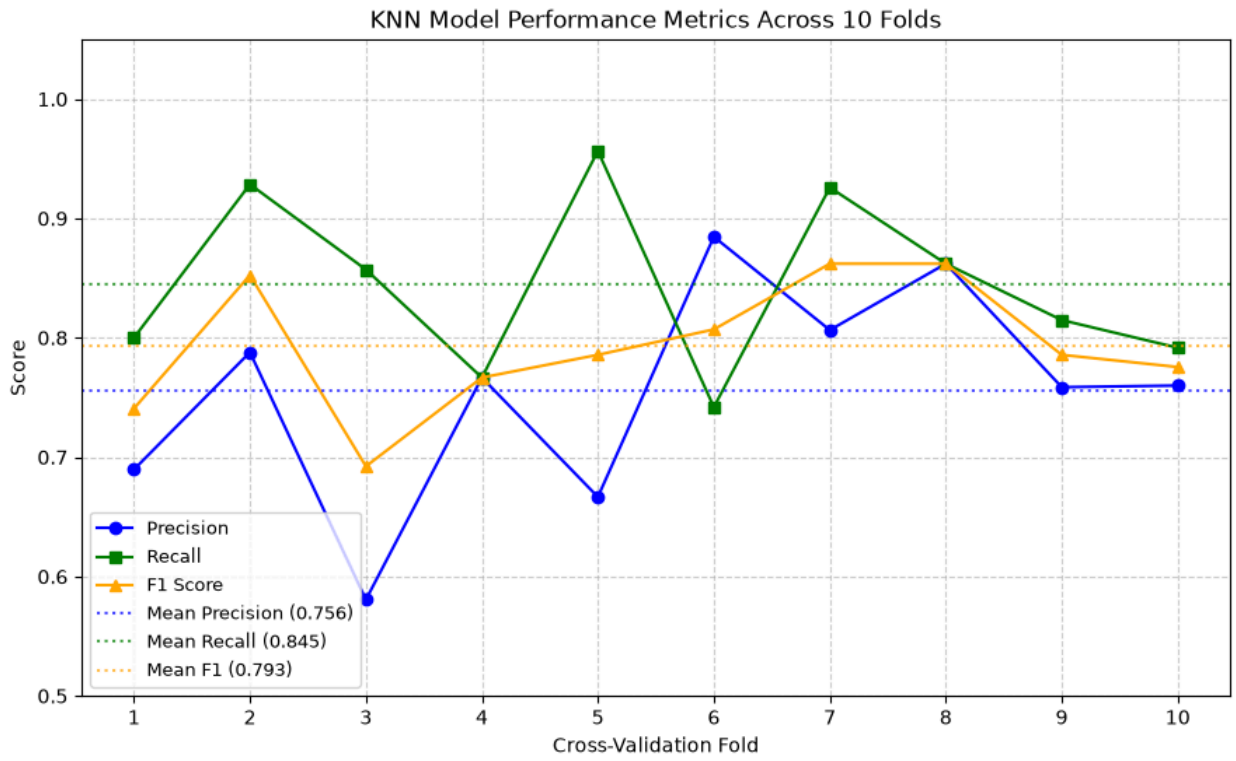
Dataset

We got this dataset from https://archive.ics.uci.edu/dataset/320/student+performance and it is called Student Performance. The dataset was donated to the website on 11/26/2014 and all of the data collected was from two secondary education Portuguese schools. The data was split into two CSV files. Each record has student attributes like absences, sex, studytime, and a number of past class failures. These were the attributes we trained the model on. The first CSV was Portuguese class, and the second CSV was math class. We chose to use the math class CSV as our dataset. This dataset had 395 students in it which we think is a good amount for the type of analysis we wanted to do. To clean this dataset we first removed all of the quotation marks so we would not run into any type error problems. Then we selected the columns we wanted to use as are attributes and named them selected_attributes. The last thing we needed to do was turn the Portuguese 1-20 grading scale into a pass and fail integer value. To do so we made anyone lower than 10 a fail and everything else a pass.

Results

We think the results we came up with are interesting but for some basic understanding of the data set it is important to keep in mind that 2 Portuguese schools do not represent every school in Portugal and most definitely do not represent schools outside of Portugal with different education structures and grading systems. Out of the dataset only 67.1% of students passed math class which seemed very low at first glance but after looking more into it that is about right if not possibly higher than the average school. I think the low pass rate can be attributed to different schooling structures than that of the USA.

We had a mean precision of .756, a mean recall of .845, and a mean F1 score of .793. A precision score of .756 means that about 75.6% of the time the model correctly predicts if a student passed or failed the math class. If we had a model that predicted pass every time the precision would be about 67.1% which makes me think that the model is correctly using the given attributes to predict the right outcome more often. A 84.5% recall means that the model correctly predicts 84.5% of the passing students and falsely says that 15.5% of passing students did not pass based on the sample we are testing against. The higher recall than precision shows that the model has an optimistic outlook, meaning it is more likely to say pass than fail when the data is not clear from the given attributes. Overall, I think this shows that a model can somewhat accurately predict student outcomes based on individual attributes like absences, sex, studytime, and number of past class failures. If the dataset was bigger this would allow for more attributes and neighbors increasing the accuracy.



Dataset Breakdown — Passing: 265 | Failing: 130 | Pass Rate: 67.1%